

MINI- REVIEW

Minireviews provides an opportunity to summarize existing knowledge of selected ecological areas, with special emphasis on current topics where rapid and significant advances are occurring. Reviews should be concise and not too wide-ranging. All key references should be cited. A summary is required.

Fishing, trophic cascades, and the structure of algal assemblages: evaluation of an old but untested paradigm

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Removal of important predators by fishing can result in trophic cascades and indirect effects on marine benthic communities. Indirect effects are especially evident when prey populations released from predation by fishing have the ability to modify entire benthic communities, as do sea urchins. Sea urchins have been shown to dramatically alter the underwater landscape by grazing, by converting stands of large erect algae into coralline barrens. In the western Mediterranean, a recent extension of coralline barrens into areas formerly dominated by erect algal assemblages has been attributed to release of predation on sea urchins by overfishing. Most suggestions concerning the transition from erect algal assemblages to coralline barrens, however, have been speculative, and little descriptive and experimental work has been carried out to verify the hypothesis that fish predation on sea urchins (and its subsequent release by overfishing) drives this transition. Here we critically review the literature concerning the effect of fishing on sea urchin populations and its subsequent maintenance of different algal assemblages in the Mediterranean. The extant data cannot refute the “fishes as important predators” model, but we argue that other processes (recruitment, pollution, disease, large-scale oceanographic events, sea urchin harvesting, food subsidies, and availability of shelters) may also be important in regulating the structure of Mediterranean algal assemblages.

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Fishing has direct effects on target species populations, reducing their abundance and in extreme cases causing their local extinction. Indirect effects of fishing eventually consist in trophic cascades after removal of top or high level predators. Such indirect effects are especially evident when prey species are able to modify entire benthic communities, as sea urchins do by grazing on

algal assemblages. Sea urchins, when at high densities, remove large erect algae and induce the formation of coralline barrens (Lawrence 1975). Stands of large erect algae and coralline barrens represent two of the most abundant algal assemblages in temperate shallow rocky habitats (e.g. Schiel and Foster 1986, Choat and Ayling 1987). It has been suggested that the alternation be-

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tween these two algal assemblages is due to fishing level, but there is controversy about the stability of algal assemblages and the processes leading to their establishment (Foster and Schiel 1988, Elner and Vadas 1990).

In the 1970s widespread destruction of kelp beds by sea urchins in the northwest Atlantic (Mann and Breen 1972, Breen and Mann 1976) resulted in the creation of a well-known ecological paradigm: the lobster–sea urchin–kelp interaction. Sea urchin population explosions that caused the formation of extensive coralline barrens along the coasts of Nova Scotia were attributed to the loss of a keystone predator (American lobster) by overfishing (see review by Elner and Vadas 1990). Although considerable research appeared to support this hypothesis, many of these studies relied on weak inference, and thus the model seems to be unreliable (e.g. Miller 1985, Elner and Vadas 1990).

The role of keystone predators was later extended to explain sea otter–sea urchin–kelp interactions in the northeast Pacific. Sea otter predation controls sea urchin populations, and therefore promotes kelp forests in areas inhabited by sea otters, whereas areas lacking sea otters are deforested by sea urchin grazing (e.g. VanBlaricom and Estes 1988). However, it has been argued that sea otters are not the only important factor regulating kelp forests (Foster and Schiel 1988). Other factors, such as other sea urchin predators (fishes and lobsters; Tegner and Dayton 1981, Cowen 1983, Tegner and Levin 1983), sea urchin recruitment variability (Estes and Duggins 1995) and physical disturbances (Dayton and Tegner 1984, Dayton et al. 1992) may also play important roles in regulating kelp forest structure. Experience obtained from the history of this often controversial paradigm should help in dealing with similar problems in other marine habitats.

In the Mediterranean rocky infralittoral, two major algal assemblages, corresponding to types of vegetation, have been characterized: coralline barrens and erect algal assemblages (Verlaque 1987a). These distinctions correspond to architectural types, and species composition can vary between regions. This is a simplification of Mediterranean algal assemblage structure, since coralline barrens and erect algal assemblages can rather be considered as terminal evolutive states within a continuum of perturbed and successional states. For instance, a unique turf algal assemblage caused by herbivorous fish grazing has been described for the Mediterranean (Verlaque 1990, Sala and Boudouresque 1997). Verlaque (1987a) also described a bush-like algal assemblage immediately after sea urchins remove large erect algae.

In some Mediterranean areas, erect algal assemblages have been replaced by coralline barrens (Neill and Larkum 1965, Larkum et al. 1967, Solazzi 1968, Giaccone 1971, Ghirardelli et al. 1974, Katzmann 1974, Edwards et al. 1975, Gros 1978, Harmelin et al. 1981,

Verlaque 1987a). Transition from erect algal assemblages to coralline barrens is recognized to result from intense grazing activity of benthic herbivores, especially sea urchins (Verlaque 1987b). The exclusion of erect algae is mainly due to the sea urchin *Paracentrotus lividus* (Lamarck) (Kempf 1962, Neill and Larkum 1965, Mastaller 1974, Stirn et al. 1974, Nédélec 1982, Vukovic 1982, Verlaque 1984, 1987a, b, Sala 1996). Another sea urchin, *Arbacia lixula* (L.), has been identified as a browser, feeding on encrusting and turf algae (Kempf 1962, Bergin 1987, Frantzis et al. 1988). *A. lixula*, together with limpets, may completely denude areas previously cleared of large erect algae by *P. lividus* (Verlaque 1987a). Transition from coralline barrens back to erect algal assemblages is possible when sea urchins are eliminated or their populations strongly reduced, as has been shown by both experimental and natural removal (Kempf 1962, Mastaller 1974, Nédélec 1982, Verlaque 1987a).

Why have many erect algal assemblages been transformed into coralline barrens in western Mediterranean areas? Since it has been demonstrated that coralline barrens are due to the intense grazing activity of sea urchins, the key to understanding the transition from erect algal communities to coralline barrens is the regulation of sea urchin populations (Fig. 1). Strong fluctuations in *P. lividus* populations have been attributed to disease, highly variable recruitment events, loss of major predators, environmental disturbance, pollution, and harvesting (Table 1). Nine out of seventeen studies postulated predation as a major factor controlling *P. lividus* populations. Most provided evidence of an increase in *P. lividus* populations and hypothesized that this was caused by the reduction of predator abundance by fishing ("fishing" will be hereafter referred to as fishing sea urchin predators, whereas "harvesting" will refer to sea urchin fisheries). Predator (mainly fish and large decapods) control of sea urchin populations implies that areas inhabited by large populations of predators would support widely distributed erect algal assemblages, whereas areas lacking predators would be cleared by sea urchin grazing and mainly composed of coralline barrens. Some of the evidence supporting this hypothesis, however, is completely speculative (Stirn et al. 1974, Savy 1987, Verlaque 1987b). Other studies are based on comparisons between protected and unprotected areas which present obvious differences in the abundance of *P. lividus* predators (Boudouresque et al. 1992, San Martín 1995). One such study presents both circumstantial (comparison between protected and unprotected areas) and experimental evidence (tethering experiments conducted within and outside a protected area) on the effects of fishing on fluctuations of sea urchin densities in the Western Mediterranean (Sala and Zabala 1996). However, to date no long-term study on the relationship between predator depletion and explosion of sea urchin populations has been conducted.

Table 1. Studies in which changes of *P. lividus* population density in the Mediterranean has been attributed to one of the following processes: disease, recruitment, predation, environmental disturbance, pollution and harvesting. The kind of predators and data which supported the hypothesis are included.

Study	Process involved	Predators	Data collection method
Allain (1972)	Environmental disturbance	—	Literature surveys
Stirn et al. (1974)	Predation	Crustaceans, fish ^a	Field surveys
Boudouresque et al. (1981)	Diseases	—	Field surveys
Harmelin et al. (1980)	Diseases	—	Field surveys
Harmelin et al. (1981)	Pollution	—	Field surveys
Azzolina (1987, 1988)	Diseases	—	Field surveys
Gras (1987)	Harvesting	Man	Field surveys
Regis (1987)	Pollution	—	Literature surveys
Savy (1987)	Predation	Crustaceans, fishes, asteroids, birds ^a	Literature surveys
Verlaque (1987b)	Predation, recruitment	Crustaceans, fishes ^a	Field surveys, literature surveys
Boudouresque et al. (1990)	Diseases, predation, recruitment	Undetermined	Literature survey
Boudouresque et al. (1992)	Predation	Fishes ^a	Field surveys
San Martín (1995)	Predation	Fishes ^b	Field surveys
Sala and Zabala (1996)	Predation, recruitment	Fishes ^b	Field surveys, field experiments
Sala (1997)	Predation	Fishes ^b	Field experiments
McClanahan and Sala (1997)	Predation	Fishes ^a	Simulation model

^a Predators have not been observed in these studies. The identification of predators is speculative, based in indirect evidence (such as stomach content analyses) or based on literature surveys.

^b Predators have been observed or identified from bites in sea urchin tests.

In this paper we analyse the consequences of fishing on sea urchin populations and algal assemblage structure in Mediterranean infralittoral by critically reviewing existing data, developing hypotheses and identifying areas for future research. Essentially, to explain the shift from erect algae assemblages to coralline barrens, we need to consider information regarding: (1) the identification of major sea urchin predators, (2) the effect of fishing on the populations of such predators, (3) the effect of predators on sea urchin populations, (4) the role of other factors in structuring sea urchin populations.

Predators and their effects on sea urchin populations

Present knowledge about sea urchin predators in the western Mediterranean is summarized in Table 2. Surprisingly, only one of seven studies claiming overfishing of predators as a major cause of *Paracentrotus lividus* population increase reliably identified them (mainly fishes, although predatory species could not be properly identified; Sala and Zabala 1996). Although 32 species have been observed as sea urchin consumers, only 17 have been identified as predators.

Crustaceans

The first group of known large predators of sea urchins is large crustaceans. Although some authors consider

large decapods as important predators of sea urchins (Stirn et al. 1974, Savy 1987, Verlaque 1987b), only three species have been identified as successful predators in the Mediterranean (Table 2). The crab *Eriphia spinifrons* inhabits the shallow infralittoral zone. The spider crab *Maja squinado* inhabits bottoms between 20 and 50 m depth in winter, and moves up to shallower habitats with erect algae in spring (Riedl 1986). Although these two species frequent the infralittoral zone, it cannot be stated that they are important predators, as there is a total lack of data concerning their predation behaviour or their predation rates on sea urchins. Spiny lobsters, *Palinurus vulgaris*, and the sea urchin *Paracentrotus lividus* have almost non-overlapping distributions, with sea urchins inhabiting shallower waters and lobsters inhabiting deeper communities. Furthermore, predation by lobsters on sea urchins has never been observed in the field. On the other hand, although the density of fishing-vulnerable decapods appears to be low in fished areas (pers. obs.), there are no quantitative data on their overall densities in the Mediterranean. Still, the modest amount of data on crustaceans does not suggest that these are important sea urchin predators.

Asteroids

The second group of large sea urchin predators is the asteroids. The spiny-starfish *Marthasterias glacialis* is a known predator of adult sea urchins, mostly *Paracentrotus lividus* (Vasserot 1964, Dance and Savy 1987).

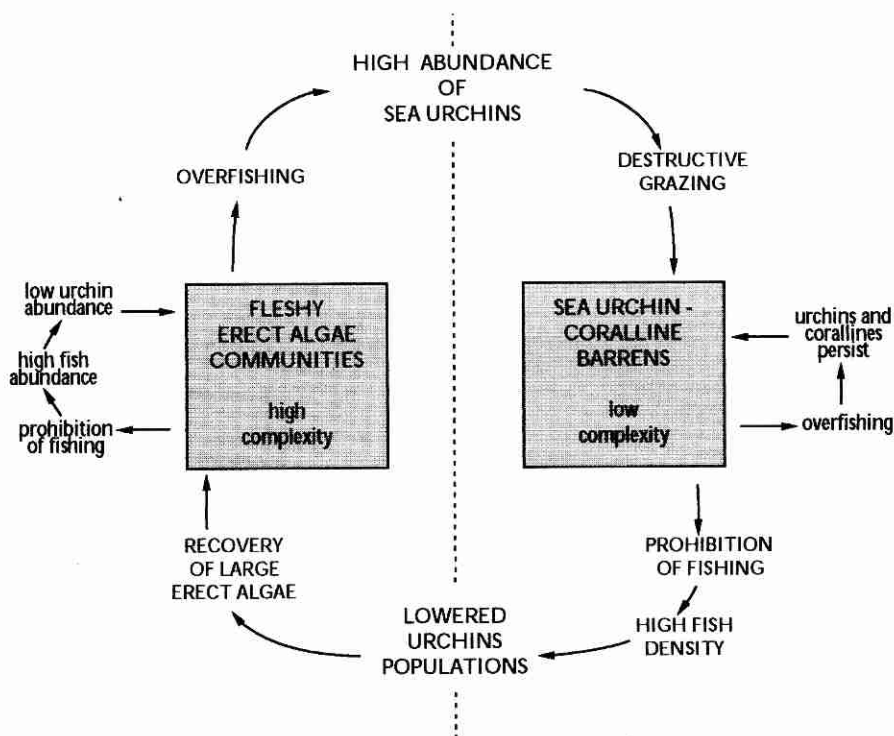


Fig. 1. Model of dynamics of western Mediterranean rocky infralittoral algal assemblages: terminal evolutive states and transition processes depending on fishing level.

Dance and Savy (1987) conducted an experimental field study to assess the impact of *M. glacialis* predation on *P. lividus* populations within a seagrass (*Posidonia oceanica* (L.) Delile) bed, and estimated that the starfish would be able to eat 10% of a *P. lividus* adult population annually, at densities of 0.07 *Marthasterias* m⁻² and 2.3 *P. lividus* m⁻². The densities of sea urchins used in this experiment, however, were low relative to other Mediterranean areas (e.g., 16 m⁻² in the Catalan coast; Sala and Zabala 1996). Furthermore, Dance and Savy (1987) experimentally cleared all fauna within 4-m² cages and introduced only *M. glacialis* and their prey *P. lividus*, with no alternative choice of prey, which suggests that the postulated effect of *Marthasterias* predation on *P. lividus* may be important under some circumstances, but it is not generalizable. Since *M. glacialis* is not subjected to fishing, and it is not a known prey item for Mediterranean littoral fishes (Khoury 1987, Sala 1996), this species is not included in the hypothesis of overfishing as major cause of sea urchin population increases.

Fishes

The most abundant and diverse group of sea urchin predators is the fishes (Savy 1987, Sala 1997). Most reports of sea urchin consumption are based on analy-

sis of fish stomach contents, which generally do not allow determination of whether a species is an active predator, or just a scavenger (a critical distinction in clarifying whether fishing affects sea urchin populations). Furthermore, sea urchins are usually pooled within an 'echinoids' group in fish feeding studies, without any identification of sea urchin species. Although 25 species of Mediterranean fish were identified as sea urchin consumers, only nine species were identified as predators from the analysis of stomach contents (small urchins were found entire within stomachs) (Table 2). A recent study based on observations of fish behaviour identified six fishes as *Paracentrotus lividus* predators, although the number of predators on juvenile sea urchins may be higher (Sala 1997). While further studies must be performed, we can state that the following fish species are successful sea urchin predators: the Sparidae *Diplodus sargus*, *Diplodus vulgaris*, *Sparus aurata*, the Labridae *Coris julis*, *Labrus merula*, *L. viridis*, *Symphodus roissali*, *S. tinca*, *Thalassoma pavo*, and the Gobiidae *Gobius fallax* and *Gobius paganellus* (Quignard 1966, Khoury 1987, Sala 1996, 1997).

As large-scale experimental removals of predatory fishes and subsequent monitoring of sea urchin populations are difficult for ethical reasons, the hypotheses concerning the role of fishing on sea urchin populations were tested a posteriori by comparing populations in protected (unfished) and unprotected (fished) areas. The

Table 2. Consumers of sea urchins in the Mediterranean. Species which eat sea urchins have been pooled into two feeding categories: predators (Pred: species which bite and break a sea urchin test) and scavengers (Scav: species which bite an opened sea urchin test or eat fragments of gonads of an opened test). Undet.: undetermined species. *P. lividus* = *Paracentrotus lividus*, *A. lixula* = *Arbacia lixula*, *S. granularis* = *Sphaerechinus granularis*, *P. microtuberculatus* = *Psammechinus microtuberculatus*, *E. acutus* = *Echinus acutus*, *E. melo* = *Echinus melo*.

Consumer species	Sea urchin species	Feeding category			Data collection	Reference
		Pred.	Scav.	Undet.		
Decapod crustaceans						
<i>Eriphia spinifrons</i> (Forsk.)	<i>P. lividus</i>	+			Stomach content analysis	Stirn et al. 1974
<i>Maia squinado</i> (Herbst)	<i>P. lividus</i>	+			Stomach content analysis	Stirn et al. 1974
<i>Palinurus vulgaris</i> (Fabr.)	<i>P. lividus</i>	+			Stomach content analysis	Campillo and Amadei 1978
	<i>P. lividus</i> , <i>A. lixula</i>	+			Aquarium experiment	Vasserot 1964
Asteroids						
<i>Marthasterias glacialis</i> (L.)	<i>P. lividus</i>	+			Field experiment	Dance and Savy 1987
	<i>P. lividus</i> , <i>A. lixula</i>	+			Aquarium experiment	Vasserot 1964
<i>Luidia ciliaris</i> (Philippi)	<i>P. lividus</i> , <i>A. lixula</i>	+			Aquarium experiment	Vasserot 1964
<i>Astropecten auranciaceus</i> (L.)	<i>P. lividus</i>	+			Stomach content analysis	Masse 1975
Serranid fishes						
<i>Serranus cabrilla</i> (L.)	<i>P. lividus</i>		+		Field observations	Sala 1997
Sparid fishes						
<i>Diplodus annularis</i> (Rafinesque)	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Diplodus cervinus</i> (Lowe)	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Diplodus sargus</i> (L.)	<i>P. lividus</i>			+	Stomach content analysis	Rosecchi 1985
	<i>P. lividus</i>	+			Stomach content analysis	Sala 1996
		+	+		Field observations	Sala 1997
<i>Diplodus vulgaris</i> (E. G. Saint-Hilaire)	<i>P. lividus</i>			+	Stomach content analysis	Rosecchi 1985
	<i>P. lividus</i>	+			Stomach content analysis	Sala 1996
		+	+		Field observations	Sala 1997
	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
<i>Oblada melanura</i> (L.)	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Sparus aurata</i> L.	<i>P. lividus</i>	+			Field observations	Sala 1997
<i>Spondyllosoma cantharus</i> (L.)	<i>P. lividus</i>		+		Field observations	Sala 1997
Pomacentrid fishes						
<i>Chromis chromis</i> (L.)	<i>P. lividus</i>		+		Field observations	Sala 1997
Labrid fishes						
<i>Coris julis</i> (L.)	Undetermined			+	Stomach content analysis	Quignard 1966
	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
	<i>P. lividus</i>	+			Stomach content analysis	Sala 1996
		+	+		Field observations	Sala 1997
<i>Labrus merula</i> (L.)	<i>P. lividus</i> , <i>S. granularis</i> , <i>P. microtuberculatus</i>	+			Stomach content analysis	Quignard 1966

Table 2. (Continued)

Consumer species	Sea urchin species	Feeding category			Data collection	Reference
		Pred.	Scav.	Undet.		
	<i>P. lividus</i>	+	+		Field observations	Sala 1997
<i>Labrus viridis</i> (L.)	<i>P. lividus</i> , <i>S. granulatus</i> , <i>E. acutus</i> , <i>E. melo</i>	+			Stomach content analysis	Quignard 1966
	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
<i>Symphodus cinereus</i> (Bonnaterre)	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
<i>Symphodus mediterraneus</i> (L.)	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983, Khoury 1987
<i>Symphodus melanocercus</i> (Risso)	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Symphodus ocellatus</i> (Forsskal)	Undetermined			+	Stomach content analysis	Khoury 1987
	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Symphodus roissali</i> (Risso)	<i>P. lividus</i> , <i>P. microtuberculatus</i>	+			Stomach content analysis	Quignard 1966
	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Symphodus rostratus</i> (Bloch)	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983, Khoury 1987
<i>Symphodus tinca</i> (L.)	<i>P. lividus</i> , <i>S. granulatus</i> , <i>P. microtuberculatus</i>	+			Stomach content analysis	Quignard 1966
	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Thalassoma pavo</i> (L.)	<i>P. lividus</i>	+	+		Field observations	Sala 1997
Gobiid fishes						
<i>Deltentosteus quadrimaculatus</i> (Valenciennes)	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
<i>Gobius cruentatus</i> Gmelin	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983
<i>Gobius fallax</i> Serato	Undetermined	+			Stomach content analysis	Khoury 1987
<i>Gobius paganellus</i> L.	Undetermined	+			Stomach content analysis	Khoury 1987
Blennioid fishes						
<i>Parablennius rouxi</i> (Cocco)	<i>P. lividus</i>		+		Field observations	Sala 1997
<i>Parablennius tentacularis</i> (Brünnich)	Undetermined			+	Stomach content analysis	Bell and Harmelin-Vivien 1983

comparison of fish assemblages inside and outside marine reserves have shown similar results whatever their geographic localization in the Mediterranean: species threatened by fishing exhibit higher abundance and larger size within the protected areas (Bell 1983, Garcia-Rubies and Zabala 1990, Francour 1991, 1994, Bayle-Sempere and Ramos-Espla 1993, Harmelin et al. 1995, Garcia-Rubies 1997). Differences observed between protected and unprotected areas have been hypothesized to be due mainly to differences in fishing

intensity. These a posteriori analyses are problematic, however, because of untested factors (mainly historical), which may confound simple interpretation of observed patterns.

Comparison of protected and unprotected areas showed that sea urchin density is lower inside than outside protected areas (Boudouresque et al. 1992, Ramos 1992, San Martín 1995, Sala and Zabala 1996), presumably because of higher sea urchin predation rates within the protected areas. However, circumstan-

tial evidence is not sufficient to confirm these hypotheses. Sala and Zabala (1996) carried out an experiment to test whether sea urchins are subjected to a higher predation pressure within protected areas. They tethered *Paracentrotus lividus* of different size classes and observed that predation rate was nearly five times higher inside than outside the protected area. The survival time for sea urchins < 3 cm was 0.7 days in the protected area, although there was no refuge for these small sea urchins during the experiment. They also demonstrated that 100% of predation was due to fishes within the reserve, whereas 43% was due to gastropods in unprotected areas.

It has been suggested that there are two important trophic guilds, rather than a single keystone species, in the western Mediterranean infralittoral: (1) predators of adult and juvenile sea urchins, including mainly *Diplodus sargus* and *Diplodus vulgaris*, and (2) predators of juvenile sea urchins, mainly composed of labrids, particularly *Coris julis*, and gobiids (Sala 1997). These species, common in most infralittoral rocky bottoms, are greatly threatened by fishing. Further, Harmelin et al. (1995) proposed two target species groups depending on the kind of fishing: (1) type A target species, including *Diplodus* spp., particularly threatened by spearfishing, and (2) type B target species, including *C. julis*, particularly threatened by angling. The mean size of all target species is significantly lower within unprotected areas and a positive relationship between the individual size of sea urchin prey and the size of predatory fish was also observed (Sala 1997). This suggests that the reduction in mean size of fish in unprotected areas may result in a smaller escape size for sea urchins, which may release sea urchins from control by fish. All evidence appears to be consistent and suggests that intense fishing pressure on sparids and labrids may have important consequences on sea urchin populations.

Processes other than fish predation that affect sea urchin population densities

Although fish predation appears to be important, we argue that there are other factors controlling sea urchin populations in the Mediterranean. In this section we review factors that can potentially affect sea urchin densities.

Recruitment

Recruitment has been suggested as an important factor in determining sea urchin abundance (Tegner and Dayton 1981, Pearse and Hines 1987, Hart and Scheibling 1988, Rowley 1989, Guillou and Michel 1993), and *Paracentrotus lividus* population structure in particular

(Verlaque 1987b, Azzolina 1988, Lozano et al. 1995, Turón et al. 1995, Sala and Zabala 1996). Recruitment of *P. lividus* occurs every year, but recruitment rates exhibit strong interannual variability (López et al. 1996). Intense recruitment episodes result in increased sea urchin abundance and multimodal size distribution (Underwood and Fairweather 1989, Sala and Zabala 1996). Therefore, it could be suggested that high recruitment events may greatly contribute to the increase of sea urchin populations and, consequently, to the development of coralline barrens (as suggested to central California, Watanabe and Harrold 1991). Conversely, low recruitment might be sufficient to maintain sea urchin densities in a situation of low adult mortality (where predator density is low and no other significant disturbances exist), thus helping to maintain extant coralline barrens.

It has been suggested, however, that high abundance of predatory fish dampens large fluctuations of *Paracentrotus lividus* populations caused by recruitment variability (Sala and Zabala 1996). This argument could explain why coralline barrens are rare within protected areas and, in turn, suggests that high recruitment rates would induce sea urchin increases only in areas where predatory fish are scarce. Nevertheless, other factors, like high variation in recruitment at a spatial scale of thousand of metres (Ebert 1983, Keesing et al. 1993), also observed by Sala and Zabala (1996), could contribute to the differences observed. Therefore, we must be cautious of making conclusions until long-term monitoring of sea urchin recruitment is undertaken, and consequent effects on adult sea urchin populations, in comparable areas subjected to different degrees of fishing, are examined.

Availability of shelter

Spatial heterogeneity may influence major processes in marine benthos, such as predation and herbivory (Menge 1976, Wiens 1976, Kosala and Pickett 1991). The availability of shelters is a very important factor in determining the distribution and exposure of sea urchins to predators (Vance and Schmitt 1979, Andrew 1993). The location of juvenile sea urchins in shelters (within crevices, beneath boulders, within dense algal populations) in the Mediterranean rocky infralittoral is a widespread pattern, independent of fish predation (Kempf 1962, Gamble 1966, Zavodnik 1967, 1976, Verlaque 1987b, Sala and Zabala 1996). A general size-dependent pattern of sea urchin exposure to predators has been observed: small individuals (generally ≤ 3 cm in diameter) are almost always sheltered within crevices, beneath boulders or even beneath larger individuals, whereas larger individuals are usually exposed to predatory fish (Sala and Zabala 1996). Sala and Zabala (1996) also observed that the higher the predatory fish

abundance, the lower the proportion of exposed sea urchins. The escape in size in *Paracentrotus lividus* was attained between 3 and 5 cm test diameter, depending on the mean size of predatory fish present (Sala 1997). These data suggest that there is a particularly dangerous phase between the exit of shelters and attaining escape size. Using these data, we can infer that a high availability of shelters may allow the survival of high numbers of sea urchin recruits after a strong recruitment episode. Once these small individuals have attained the escape in size they will be able to leave shelters without apparent predation risk and denude erect algae. However, we do not know whether bimodal patterns are due to faster growth once urchins leave shelters (*P. lividus* shows density-dependent growth rate; Grosjean et al. 1996).

The availability of shelter may also induce the formation of coralline barrens. In temperate Australia, an experimental study showed that the sea urchin *Centrostephanus rodgersii* (Agassiz) causes the transition from kelp habitat to barrens habitat if shelter is provided (Andrew 1993). In this experiment, sea urchins recruited irrespective of the availability of shelter, but densities declined in patches where shelter was minimized. In the Medes Islands marine reserve (NE Spain), where most urchins are restricted to shelters (Sala and Zabala 1996), the effect of the availability of shelter is striking. Halos of coralline barrens were observed around the base of infralittoral boulders, whereas the top of boulders, away from the main shelter and where urchins would be exposed to predators, was covered by erect algae (Sala 1996). However, in a nearby (~2 km) similar bottom subjected to fishing, sea urchins were distributed all over the surface of boulders, with the consequent dominance of coralline barrens even at the top of boulders. Predatory fish abundance appeared to be the cause of these changes in sea urchin behaviour. Sala (1996) suggested a "refuge distance" of 50 cm, within which *Paracentrotus lividus* graze at night and avoid diurnal predators (mainly fish). This predator avoidance strategy (escape in time) has also been observed in other Mediterranean areas (Kempf 1962, Gamble 1965; but see Dance 1987a, b). Such evidence may explain the recent observation of a coralline barren area in the Medes Islands marine reserve, where predatory fish abundance is the highest recorded in the Mediterranean (García-Rubies and Zabala 1990, Sala 1996). In this barren habitat there is a high abundance of crevices, with narrow inter-crevice distances (shorter than the sea-urchin refuge distance at this site). Sala (1996) suggested this should allow sea urchins to eliminate all erect algae within this site. This circumstantial evidence suggests that the abundance of shelters may facilitate the formation of barren areas even within sites with high abundance of predatory fish. However, no experimental study has been designed to quantify the respective roles of shelter availability and predatory fish

abundance in the formation of coralline barrens in the Mediterranean.

Oceanographic events

A twelve-fold increase in the abundance of the sea urchin *Arbacia lixula* in only one decade has been reported in the NW Mediterranean (from 8 100 m⁻² before 1985 to 100 100 m⁻² in 1992 in Corsica; Francour et al. 1994). This was attributed to the increase in the average water temperature of the western Mediterranean basin. As the increasing abundance of *A. lixula* was detected within protected areas supporting high abundance of predatory fishes (Francour et al. 1994, pers. obs.), there is no reason to link this increase to release from predation due to fishing. Much higher densities of *A. lixula*, however, were found in the Marseille region in the 1960s (> 10 m⁻²; Kempf 1962), indicating long-term changes in *A. lixula* abundance. Temperature anomalies have been also suggested to affect sea urchin recruitment and hence adult populations in other regions. Hart and Scheibling (1988) found that high seawater temperature causes rapid larval growth and development of *Strongylocentrotus droebachiensis* (Müller), with consequent increase of recruitment. They hypothesized that destructive grazing in Nova Scotia kelp assemblages was favoured by major recruitment events associated with high temperatures. Therefore, modification of the composition and abundance of herbivore assemblages by long-term environmental changes may have ecological consequences similar to those attributed to overfishing.

Pollution

Eutrophication has a clear influence on *Paracentrotus lividus* abundance (Regis 1981, Harmelin et al. 1981, Delmas and Régis 1984, Delmas 1988, Pancucci et al. 1993). Coastal zones located near sewage plants support high densities of *P. lividus* (38 m⁻²; Harmelin et al. 1981), as well as those near release of agricultural effluents (110 m⁻²; Pancucci et al. 1993), and are associated with extensive coralline barrens. Since *P. lividus* may assimilate dissolved nutrients directly from seawater (Regis 1986), sewage runoff can contribute to a population increase. Indeed, it has been suggested that *P. lividus* reproduces more actively in areas where energy derived from dissolved organic matter exceeds that produced by phyto-benthos (Regis 1979, Pancucci et al. 1993). In addition, phytoplankton growth caused by runoffs (Estrada 1979, Béthoux 1981) may also have an influence on *P. lividus* recruitment, as has been shown for other sea urchin species (Starr et al. 1990, 1993). Since the Mediterranean suffers at present from increased coastal urbanization, increasing pollution as-

sociated to this trend (Albaigés et al. 1985) should be considered when assessing the causes of recent sea urchin proliferation.

Inputs of allochthonous food subsidies

Consumer populations which are subsidized by imported allochthonous trophic resources may increase so as to suppress prey populations (Polis and Strong 1996 and references therein). For instance, kelp detritus from the sublittoral zone promotes dense intertidal limpet and sea urchin populations, with consequent denudation of erect algae (Duggins et al. 1989, Bustamante et al. 1995). Likewise, abundance of drift algae can trigger changes in sea urchin grazing behaviour, with the subsequent deforestation of a kelp forest, without changes in the abundance of sea urchins (Ebeling et al. 1985, Harrold and Reed 1985, Watanabe and Harrold 1991). In the Mediterranean rocky infralittoral, subsidies from the seagrass meadows may be very important in providing food resources to sea urchins. Drift blades of the seagrass *Posidonia oceanica* may contribute largely to the diet of the sea urchin *Paracentrotus lividus* in rocky bottoms where the seagrass is not present (up to 40% of stomach contents in some seasons; Verlaque and Nédélec 1983). Since a considerable amount of detritus of *P. oceanica* meadows are exported (average of 50% between 1 and 30 m depth; Romero et al. 1992), this spatial coupling in food resources might help to maintain sea urchin populations and coralline barrens.

Sea urchin harvesting

Sea urchins gonads are a delicacy in Asian and Mediterranean countries (Sloan 1984, Le Direac'h et al. 1987). The impact of human predation (commercial and recreational) on large-sized sea urchins could prove very important; in the Marseilles region (France), intense *P. lividus* harvesting has also caused local depletions of the urchin stocks (Gras 1987). In contrast, in other regions where they are not harvested (such as Greece), they attain such high densities that fishermen complain (Le Direac'h et al. 1987). Therefore, sea urchin population outbreaks should not be a problem in areas where urchins are harvested; on the contrary, they may be overfished, as occurred with *P. lividus* in Brittany (France), where stocks were depleted and the fishery halted (Southward and Southward 1975, Le Gall 1987).

Disease

Disease has decimated sea urchin populations from several regions, with subsequent indirect effects on benthic communities (Pearse and Hines 1987, Scheibling

and Stephenson 1984, Lessios 1988, Andrew 1991, Hagen 1992). In the Mediterranean, the only report of a sea urchin mass mortality was that of the bald sea urchin disease that occurred along the French coast in the 1980s. The impact of the disease was especially apparent at the Port-Cros National Park (Var, France), where it strongly reduced *Paracentrotus lividus* abundance (up to a 10-fold reduction from 1979 to 1983; Boudouresque et al. 1981, Azzolina 1983, 1987, 1988). The dramatic drop of *P. lividus* abundance coupled with intense harvesting may have caused the collapse of some populations, suggesting an important role of such factors which counteract the putative impact of overfishing. Unfortunately, no monitoring of algal assemblages was carried out after the disease outbreak.

Fishing intensity and algal assemblages: hypotheses that arise from evidence

Verlaque (1987b) outlined several hypotheses to explain sea urchin proliferation, the most likely being high recruitment and decrease in mortality during benthic life of urchins. In this paper we have highlighted the importance of topographic structure of the substratum, recruitment rate, pollution, foods subsidies, and predatory fish abundance as proximal causes of sea urchin abundance. Taking into account the interactions between these factors we suggest the following model to be tested in the future. There is a strong interaction between sea urchins and algae: sea urchins severely consume erect algae, maintain a low erect algae/en-crusting coralline algae ratio and induce the formation of coralline barrens. Herbivorous fishes can also decrease this ratio, although at higher levels. Regulation of sea urchin populations is hence the central issue, and is driven by balance between recruitment and mortality (Fig. 2). Recruitment may be affected by pollution and by reproductive adult stocks, which in turn are affected by harvesting. Mortality is not only due to predation (harvesting by man or fish predation), but also by disease and pollution (strong chemical pollution). Predation can significantly reduce sea urchin populations, as evidenced by the situation in marine reserves where predatory fish abundance is high. However, the topographic structure of the habitat may play a determining role on the effect of fish predation on sea urchin populations. An abundance of suitable shelters in protected areas allows sea urchins to persist, albeit in lower abundance than in fished areas, which results in a mosaic of erect algal patches and coralline barrens. Further, predation can be mediated by extreme high and low recruitment events, which in turn can be facilitated by organic pollution, hence potentially regulating sea urchin abundance in spite of high predatory fish abundance. Yet, recruitment episodes can occur independently of increasing pollution. Intense fishing

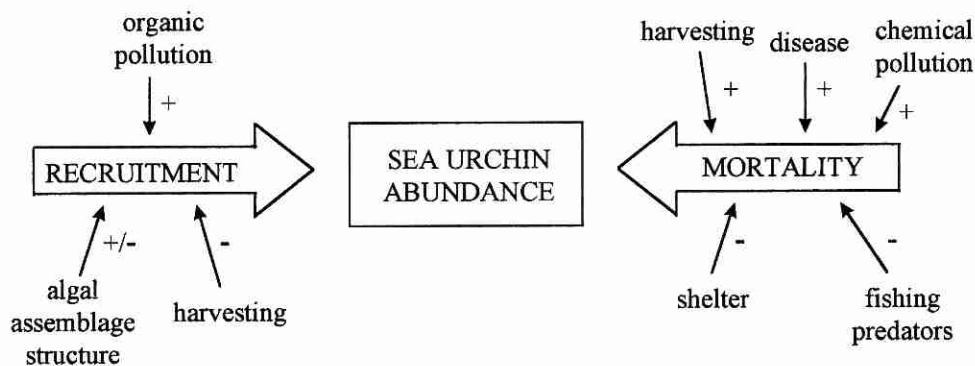


Fig. 2. Model of processes that control sea urchin abundance and factors that mediate these processes. Negative signs represent negative mediation (the factor diminishes the importance of the process), while positive signs represent positive mediation (the factor enhances the process).

decreases predatory fish abundance, but in some Mediterranean areas, harvesting of sea urchins occurs in addition to fishing. Low abundances of herbivorous fish and urchins reduce the grazing pressure on the algal assemblages, and enhance erect algal cover.

The first concern of this model is that most research from which it arises has been conducted in the western Mediterranean, mainly in the Gulf of Lions, but scarce attention has been paid to the extension of coralline barrens in other Mediterranean regions. Generalization to the whole Mediterranean sea is thus subject to caution. During recent field work in the Balearic Islands, one of us (E.S.) observed that, at several sites with intense fishing activity, sea urchins were at low densities (compared to those of the Gulf of Lions), and erect algal assemblages dominated. Furthermore, sea urchin densities were similar or even higher within a marine national park where fishing was prohibited and predatory fish very abundant (unpubl.). Harvesting of sea urchins does not occur at these sites. These observations suggest that at the Balearic Islands processes other than fish predation can limit the density of sea urchin populations. Environmental characteristics particular to each site may play an important role in determining the strength and the result of predator–urchin–algae interactions. Although fish predation may have an overall influence on the structure of Mediterranean algal assemblages, an array of processes can come into play when predation is relaxed, creating a pattern which is neither easily nor accurately predictable (Foster and Schiel 1988). Before generalizing from field data, a balance between confidence in the data and generality must be accomplished. That means that further studies should be dispersed within the entire Mediterranean, or at least at regional scales.

A second concern is that most assessments concerning predation have been based on the activity of diurnal predators (mostly fishes). However, we do not know much about the role of nocturnal predators on sea urchin populations. It has been shown that sea urchins

are preferentially nocturnal (Gamble 1965), and therefore their vulnerability would increase during nocturnal migrations because some predators also have preferential nocturnal activity (such as the spiny-starfish *Marthasterias glacialis*; Savy 1987). In addition, the cryptic behaviour of sea urchins (sheltered within crevices or beneath boulders) may reduce predation from diurnal predatory fishes (Sala and Zabala 1996). The effectiveness of refuges against other predators such as starfishes and gastropods (such as the muricid *Trunculariopsis trunculus*; Sala and Zabala 1996) is unknown.

A third concern that has received little attention is juvenile mortality (Sala and Zabala 1996). While predation on adults may be mainly by large fishes, predation on recruits (individuals smaller than 10 mm in diameter) may involve smaller fishes or invertebrates (crustaceans, gastropods and polychaetes) (Harrold et al. 1991, Sala and Zabala 1996, Sala 1997). Sala and Zabala (1996) found that recruit mortality rate did not differ significantly between areas with significant differences in predatory fish abundance. Since small invertebrate predators are not fished, the release from predation by large fishes may result in an increase in small sea urchin predators and, in turn, in an increase in sea urchin recruit mortality.

Finally, there are few long-term studies of benthic communities in the Mediterranean and no data exist concerning the stability of Mediterranean coralline barrens. To our knowledge, the longest monitoring involves echinoderms in the Port-Cros National Park (France) from 1979 to the present. Unfortunately, simultaneous monitoring of algal assemblages has not been conducted. At this point we do not know whether coralline barrens constitute an irreversible state in areas where intense fishing is carried out, or if they are part of a cyclic phenomenon. Fluctuations of sea urchin populations, as occurred in Nova Scotia, where the deforestation of kelp beds in the 1970s was followed by sea urchin disease and subsequent recovery of kelps in

the 1980s (Scheibling 1986), suggest that large-scale, long-term studies are necessary.

Predator control of sea urchins by fishes in other regions

Perhaps the best example of how overfishing causes trophic cascades and determines the structure of benthic communities is that of Kenyan coral reefs. McClanahan and co-workers have carried out several comparative and experimental studies which indicate that overfishing of sea urchin predators results in increased sea urchin density and, in turn, by erosive grazing, in degradation of the coral reef (McClanahan and Muthiga 1989, McClanahan and Kurtis 1991, McClanahan 1994, McClanahan and Mutere 1994, McClanahan et al. 1994). To assess the historical role of overfishing, they carried out experimental sea urchin removals in a recently protected area with high abundance of urchins, and found that abundant sea urchins can suppress the recovery of fish populations once fishing is reduced (McClanahan et al. 1996). Overfishing appears to have inertial effects even once fishing is stopped, thus contributing to the maintenance of sea urchin overgrazed communities.

Fishes (mainly sheephead, *Semicossyphus pulcher*) and spiny lobsters (*Panulirus interruptus*) have been suggested to control sea urchin populations in southern California kelp forests (e.g. Tegner and Dayton 1981, Tegner and Levin 1983). To test for the effect of sheephead on red sea urchin populations in a southern California kelp forest, Cowen (1983) speared all sheephead in a 12 700-m² area, and then monitored sea urchin abundance and distribution for 2 years. Red sea urchin density increased substantially (26% increase/year) and urchins occupied more exposed locations in the experimental area, while both density and distribution did not change in a nearby control area. These data provide strong evidence that overfishing can result in increase of sea urchin populations, but again, we argue that fishing is not the single most important factor, because deforestation of kelp forests by sea urchins has been noted in areas in which sheephead were present, suggesting the prevalence of other factors, such as availability of drift algae (Harrold and Reed 1985, Watanabe and Harrold 1991).

Future directions

We have discussed what components of the fish-sea urchin-algae system have been tested in the Mediterranean. Some questions have been satisfactorily answered, but most still require further effort. One descriptive and experimental study has been carried out

to determine the causes of recent increases in sea urchin abundance in the NW Mediterranean (Sala and Zabala 1996). This was a local study, however, and generalization of its results involves certain risks. Although circumstantial evidence suggests similar trends for other Mediterranean areas, strong affirmations must be preceded by long-term data sets and accurate observations, and the explanatory hypotheses and models confirmed by experimental studies.

The following questions remain to be answered: What is the current spatial extent of coralline barrens in the Mediterranean? Are coralline barrens temporally stable and/or irreversible if current conditions persist? Do coralline barrens exist in areas with low abundance of sea urchins? What are the relationships between the sea urchins *Paracentrotus lividus* and *Arbacia lixula*? Can *A. lixula* alone deforest an erect algal assemblage? Would the availability of shelter for sea urchins result in a shift in algal assemblages? Do *P. lividus* recruitment limitations exist in the Mediterranean? What is the role of allochthonous food resources in maintaining high sea urchin densities? What regulates sea urchin populations within areas with low predation pressure and high settlement rates? What is the role of nocturnal predators on sea urchin populations? Are fishes able to recover in extensive barren areas with well-established sea urchin populations? What is the role of increasing organic pollution on sea urchin population increases? Under which conditions are either biotic or environmental triggers more important? These questions could serve to ascertain the relative importance of fishing pressure as a key factor for structuring sea urchin populations and its interactions with other factors in order to establish the applicability of the proposed model. Further studies should be carried out at large spatial (10s–100s of km) and temporal (years) scales, also taking advantage of the situations provided by marine reserves.

Our aim here was to critically analyse what we know and what we need to know before overfishing in the Mediterranean can be defined as a causal factor in algal assemblage structure. Although we believe that the actual evidence, both circumstantial and experimental, does not refute the hypothesis that fishing level is important in determining the alternation between erect algal assemblages and coralline barrens in the Mediterranean, we are also aware that patterns in nature can be rarely attributed to single processes; multiple controls occur in all ecosystems (Foreman et al. 1995, Polis and Strong 1996). Therefore, we suggest that overfishing may be an important factor in determining the formation of extensive coralline barrens, but the shift from erect algal stands to coralline barrens can be triggered by other factors that deserve further attention.

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